

ELITE MATHEMATICS

Pearson Edexcel International GCSE

Linear 4MA1

Adaptive Classified

OL-T02 · Set Notation and Venn Diagrams

With Answers

Questions with full worked solutions

■ 46 pages

FAMILY MAP

Set Notation Venn Diagrams

F01	Translate and operate on written sets	7 Q 17 marks
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F02	Read or construct a two-set Venn diagram	6 Q 8 marks
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F03	Resolve a three-set Venn model	8 Q 19 marks
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Question 04

4MA1-1H Q7

ORIGINAL QUESTION

5 marks

7 $\mathcal{E} = \{23, 24, 25, 26, 27, 28, 29, 30, 31, 32, 33, 34\}$

$A = \{\text{even numbers}\}$

$B = \{23, 29, 31\}$

$C = \{\text{multiples of 3}\}$

(a) List the members of the set

(i) $B \cup C$

(1)

(ii) $A' \cap C$

(1)

(b) Is it true that $B \cap C = \emptyset$?

Tick (✓) one of the boxes below.

Yes

No

☐
☐

Give a reason for your answer.

(1)

The set D has 4 members and is such that $D \cap (A \cup C) = \emptyset$

(c) List the members of set D

(2)

Worked Solution 04

Set notation

MODULAR UNIT 1

5 marks

Union and intersection

$$B \cup C = \{23, 24, 27, 29, 30, 31, 33\}, \quad A' \cap C = \{27, 33\}.$$

Empty intersection

$$B \cap C = \emptyset, \text{ so the statement is true.}$$

Complement

$$D = (A \cup C)' = \{23, 25, 29, 31\}.$$

Final answer

$$\{23, 24, 27, 29, 30, 31, 33\}; \{27, 33\}; \text{ Yes; } D = \{23, 25, 29, 31\}$$

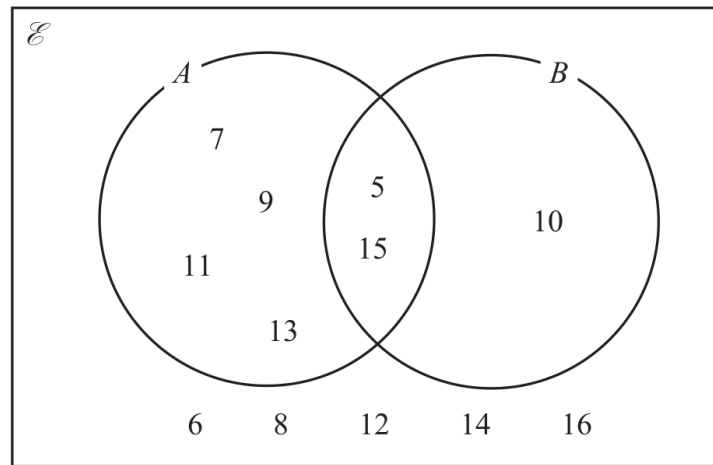
Question 01

4MA1-1H Q1 (a)

ORIGINAL QUESTION UNIT 1

1 marks

1 Here is a Venn diagram.



List the members of the set

(a) A

.....
(1)

Worked solution 01

Read members of A

MODULAR UNIT 1

1 marks

(a) – Read members of A
Read the diagram

$$A = 7, 9, 11, 13, 5, 15.$$

Final answer

$$A = \{5, 7, 9, 11, 13, 15\}$$

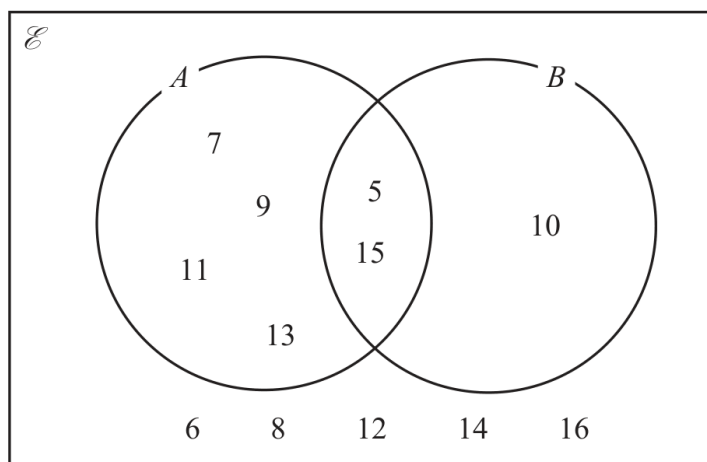
Question 02

4MA1-1H Q1 (b)

ORIGINAL QUESTION UNIT 1

1 marks

1 Here is a Venn diagram.



List the members of the set

(b) $A \cap B$

.....
(1)

Worked solution 02

Read the intersection

MODULAR UNIT 1

1 marks

(b) – Read the intersection

Read the overlap

$$A \cap B = 5, 15.$$

Final answer

$$A \cap B = \{5, 15\}$$

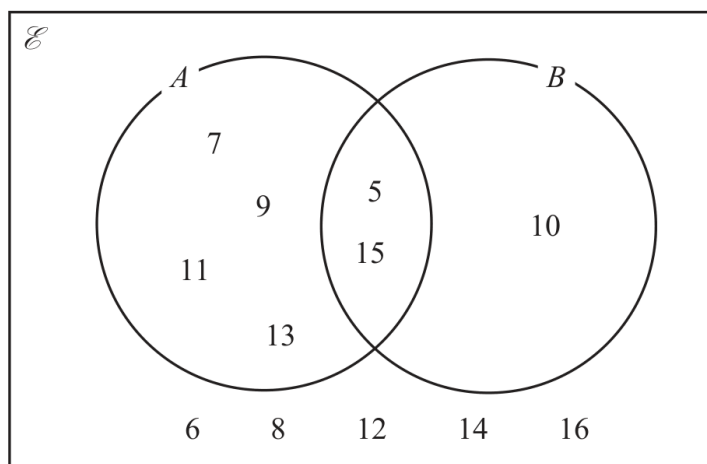
Question 03

4MA1-1H Q1 (c)

ORIGINAL QUESTION UNIT 1

1 marks

1 Here is a Venn diagram.



List the members of the set

(c) $(A \cup B)'$

(1)

Worked solution 03

Read the complement of the union

MODULAR UNIT 1

1 marks

(c) – Read the complement of the union

Read outside both circles

$$(A \cup B)' = 6, 8, 12, 14, 16.$$

Final answer

$$(A \cup B)' = \{6, 8, 12, 14, 16\}$$

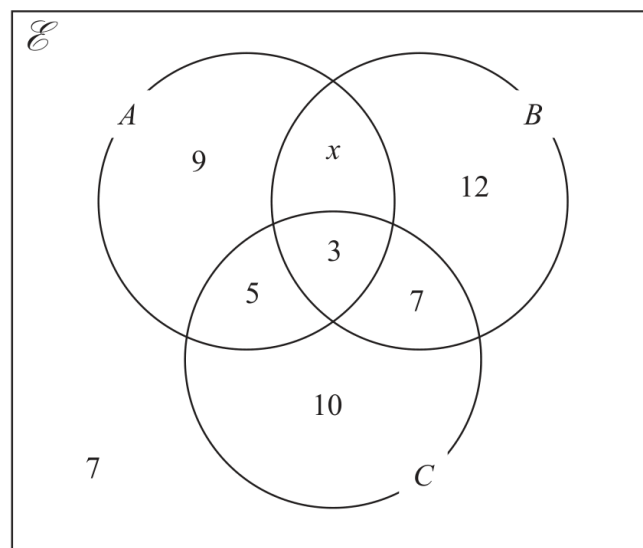
Question 09

4MA1-1H Q16

ORIGINAL QUESTION**3 marks**

16 The Venn diagram shows a universal set $\mathcal{E}$ and sets A , B and C

The numbers and the letter x represent **numbers** of elements.



Given that $n(A \cup B) = 42$

(a) find the value of x

$x = \dots\dots\dots$
(1)

(b) Find $n(A')$

$\dots\dots\dots$
(1)

(c) Find $n(B' \cap C)$

$\dots\dots\dots$
(1)

Worked Solution 09

Venn-diagram counts

MODULAR UNIT 1

3 marks

Use the union total

$$9 + 12 + x + 5 + 7 + 3 = 42 \Rightarrow x = 6.$$

Count outside A

$$n(A') = 12 + 7 + 10 + 7 = 36.$$

Count C but not B

$$n(B' \cap C) = 5 + 10 = 15.$$

Final answer

$$x = 6; \quad n(A') = 36; \quad n(B' \cap C) = 15$$

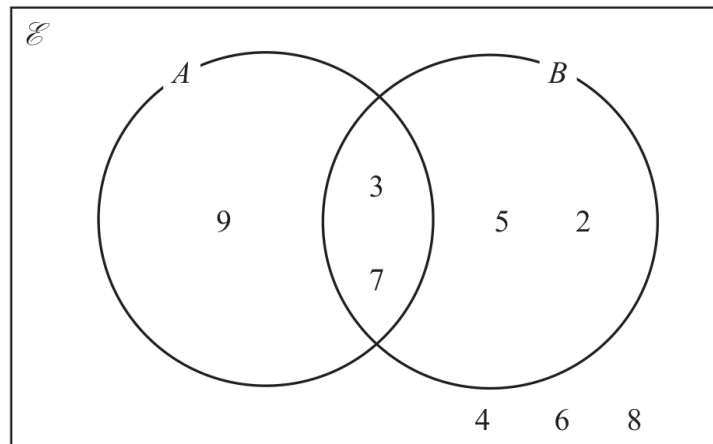
Original question 04

4MA1-1H Q4 M01

MODULAR UNIT 1

1 marks

4 Here is a Venn diagram.



(a) List the members of the set B

.....
(1)

Worked solution 04

List the members of B

MODULAR UNIT 1

1 marks

M01 – List the members of B**Read the B circle**

The elements inside B are 2,3,5,7.

Final answer

$$B = \{2, 3, 5, 7\}$$

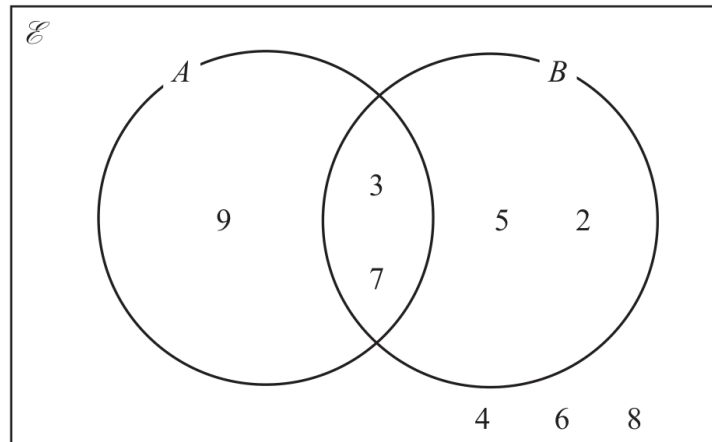
Original question 05

4MA1-1H Q4 M02

MODULAR UNIT 1

1 marks

4 Here is a Venn diagram.



(b) List the members of the set $A \cap B$

.....
(1)

Worked solution 05

List the intersection

MODULAR UNIT 1

1 marks

M02 – List the intersection**Read the overlap**

The common elements in A and B are 3 and 7.

Final answer

$$A \cap B = \{3, 7\}$$

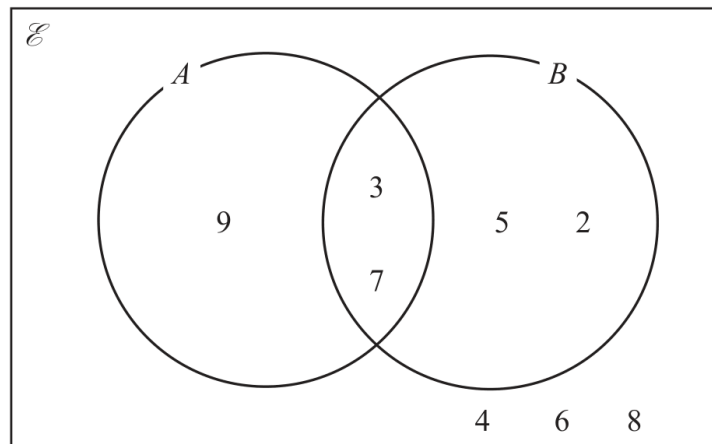
Original question 06

4MA1-1H Q4 M03

MODULAR UNIT 1

1 marks

4 Here is a Venn diagram.



(c) List the members of the set A'

.....
(1)

Worked solution 06

List the complement

MODULAR UNIT 1

1 marks

M03 – List the complement**Read outside A**

The elements in the universal set but not in A are 2,4,5,6,8.

Final answer

$$A' = \{2, 4, 5, 6, 8\}$$

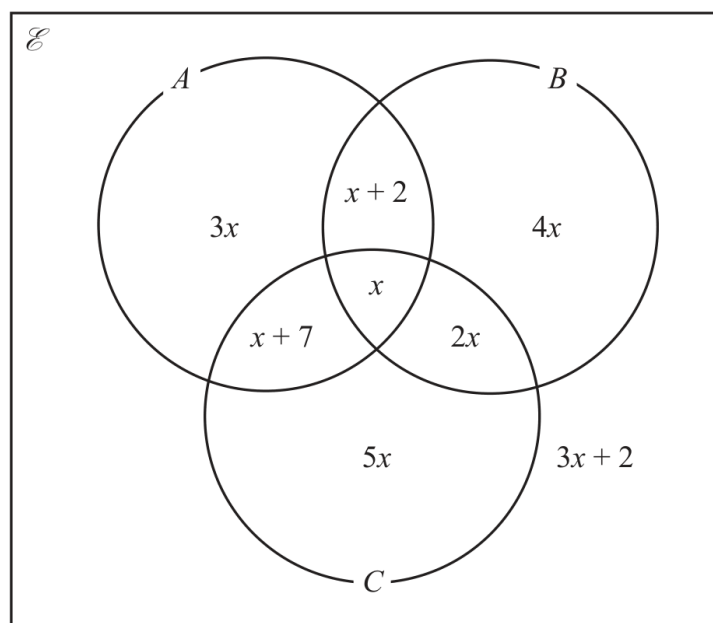
Original question 26

4MA1-2H Q21 M01

MODULAR UNIT 1

4 marks

- 21 The Venn diagram shows information about the numbers of items in set A , set B and set C , where x is an integer.



Given that $n(A \cup B)' = 26$

find $n(A' \cap C)$

$n(A' \cap C) = \dots\dots\dots$

Worked solution 26

Use a three-set Venn diagram

MODULAR UNIT 1

4 marks

M01 – Use a three-set Venn diagram**Use the complement of the union**

$$n((A \cup B)') = 5x + (3x + 2) = 26 \Rightarrow 8x = 24 \Rightarrow x = 3.$$

Select the requested regions

$$n(A' \cap C) = 5x + 2x = 7x.$$

Substitute

$$7(3) = 21.$$

Final answer

21

Original question 06

4MA1-1H Q7

ORIGINAL QUESTION**3 marks**

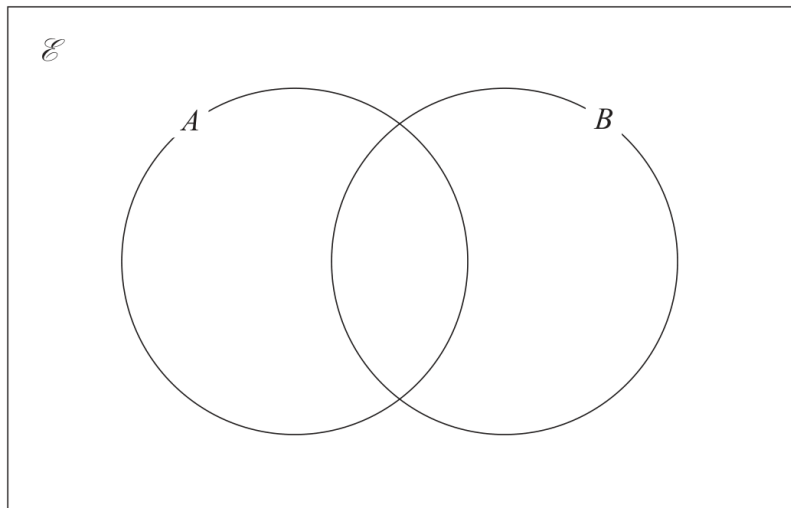
7 $\mathcal{E} = \{4, 5, 6, 7, 8, 9, 10, 11, 12, 13, 14, 15\}$

$$A \cap B = \{5, 10, 15\}$$

$$B' = \{7, 8, 9, 11, 12, 13, 14\}$$

$$A' = \{4, 6, 7, 8, 14\}$$

Complete the Venn diagram for this information.



Worked solution 06

Chapter: Set Notation & Venn Diagrams

MODULAR UNIT 1

3 marks

Q7 – Chapter: Set Notation & Venn Diagrams

Family: Read or construct a two-set Venn diagram • Variant: Complete a two-set diagram from membership data

Method

$$A \cap B = \{5, 10, 15\}.$$

Why: These elements belong to both sets.

Method

$$A \text{ only} = \{9, 11, 12, 13\}, B \text{ only} = \{4, 6\}, \text{ and outside} = \{7, 8, 14\}.$$

Why: Use the complements in the given universal set.

Method

Place the three groups in the corresponding Venn regions.

Why: Each element is used exactly once.

Final answer

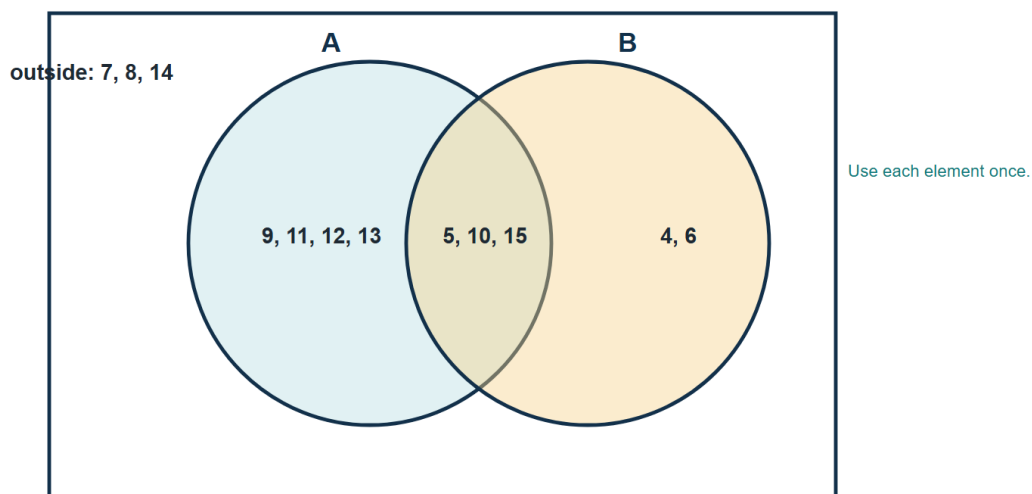
Venn regions: A only $\{9, 11, 12, 13\}$; $A \cap B$ $\{5, 10, 15\}$; B only $\{4, 6\}$; outside $\{7, 8, 14\}$

Completed diagram 06

Chapter: Set Notation & Venn Diagrams

MODULAR UNIT 1

3 marks

Q7 • Completed Venn diagram $A \cap B = \{5, 10, 15\}$; A only = $\{9, 11, 12, 13\}$; B only = $\{4, 6\}$ 

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4 $\mathcal{E} = \{20, 21, 22, 23, 24, 25, 26, 27, 28, 29\}$

$A = \{\text{odd numbers}\}$

$B = \{\text{multiples of 3}\}$

List the members of the set

(i) $A \cap B$

(1)

Worked solution

January 2021 | Question 4 | 1 marks

Family: Translate and operate on written sets • Variant: Union and intersection from listed sets

Method / working

1. Odd multiples of 3 in E are 21 and 27.

Final answer: {21, 27}

(ii) $A \cup B$

(1)

Worked solution

January 2021 | Question 4 | 1 marks

Family: Translate and operate on written sets • Variant: Union and intersection from listed sets

Method / working

1. Combine the odd numbers and multiples of 3, removing duplicates.

Final answer: {21, 23, 24, 25, 27, 29}

16 Some students were asked the following question.

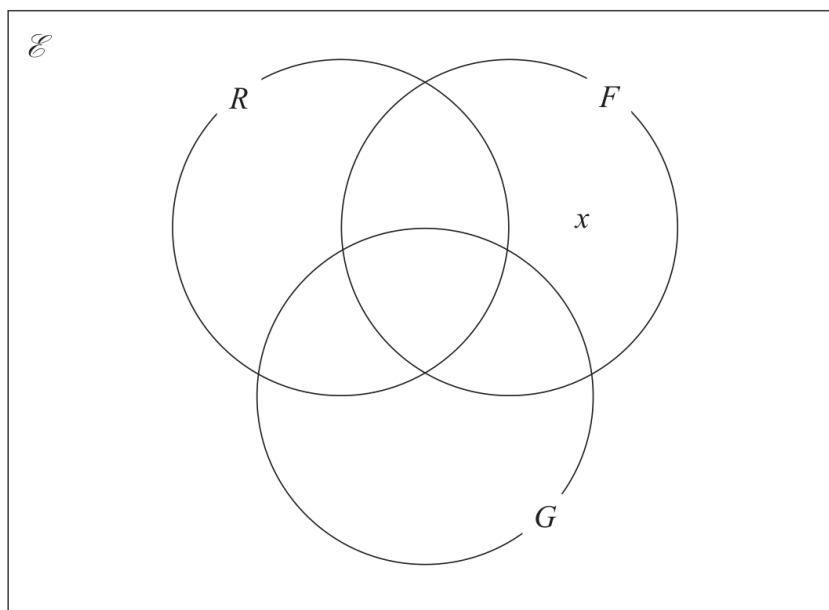
“Which of the subjects Russian (R), French (F) and German (G) do you study?”

Of these students

- 4 study all three of Russian, French and German
- 10 study Russian and French
- 13 study French and German
- 6 study Russian and German
- 24 study German
- 11 study none of the three subjects
- the number who study Russian only is twice the number who study French only.

Let x be the number of students who study French only.

(a) Show all this information on the Venn diagram, giving the number of students in each appropriate subset, in terms of x where necessary.



(3)

Given that the number of students who were asked the question was 80

(b) work out the number of these students that study Russian.

(3)

Worked solution

January 2021 | Question 16 | 3 marks

Family: Resolve a three-set Venn model • Variant: Complete or solve a practical three-set survey

Method / working

1. Place 4 in the triple intersection, $10-4=6$ in R intersection F only, $13-4=9$ in F intersection G only, and $6-4=2$ in R intersection G only.

2. Let French-only= x and Russian-only= $2x$; German-only= $24-4-9-2=9$.

3. Place 11 outside and label every region on the supplied Venn diagram.

Final answer: Completed Venn diagram with x , $2x$, 6, 2, 4, 9, 9, 11

16 Some students were asked the following question.

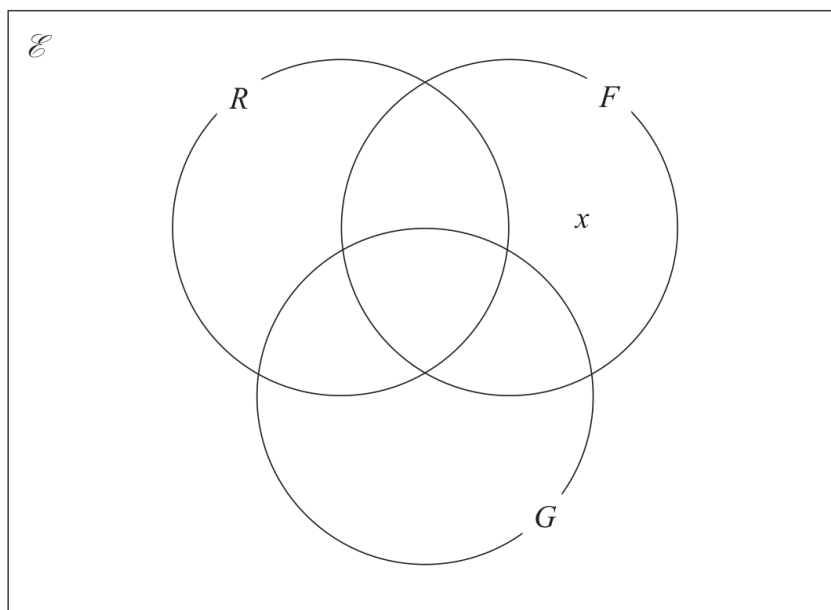
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(3)

Worked solution

January 2021 | Question 16 | 3 marks

Family: Resolve a three-set Venn model • Variant: Complete or solve a practical three-set survey

Method / working

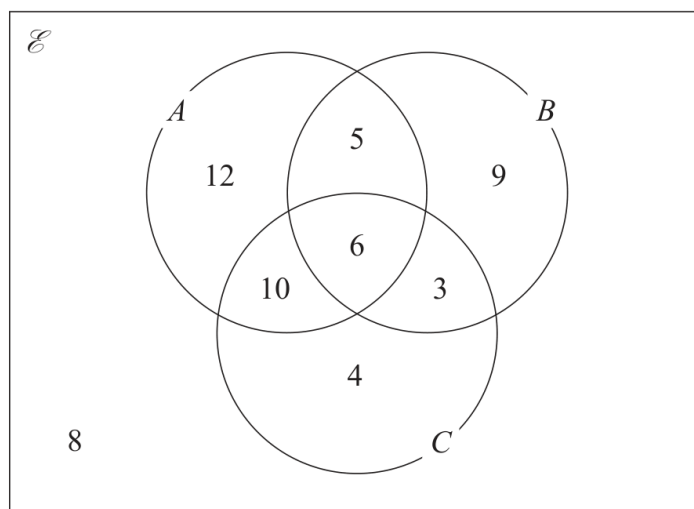
1. Use the total: $2x+6+x+2+4+9+9+11=80$.

2. Solve $3x+41=80$ to get $x=13$.

3. Russian total= $2x+6+2+4=38$.

Final answer: 38 students

19 The Venn diagram shows a universal set, $\mathcal{E}$ and sets A , B and C .



12, 5, 9, 10, 6, 3, 4 and 8 represent the **numbers** of elements.

Find

(i) $n(A \cup B)$

(1)

Worked solution

June 2021 | Question 19 | 1 marks

Family: Resolve a three-set Venn model • Variant: Read and combine three-set regions

Method / working

1. Read the required region count from the Venn diagram: 45.

Final answer: 45

(ii) $n(A' \cap B')$

(1)

Worked solution

June 2021 | Question 19 | 1 marks

Family: Resolve a three-set Venn model • Variant: Read and combine three-set regions

Method / working

1. Add the regions in $A' \cap B'$: the diagram gives 12.

Final answer: 12

(iii) $n([A \cap B] \cup C)$

(1)

Worked solution

June 2021 | Question 19 | 1 marks

Family: Resolve a three-set Venn model • Variant: Read and combine three-set regions

Method / working

1. Count $(A \cap B) \cup C$ from the labelled regions: 28.

Final answer: 28

- 4 $\mathcal{E} = \{\text{letters of the alphabet}\}$
 $B = \{\text{b, r, a, z, i, l}\}$
 $I = \{\text{i, r, e, l, a, n, d}\}$

(a) List the members of the set

(i) $B \cup I$

(ii) $B \cap I'$

(2)

Worked solution

June 2021 | Question 4 | 2 marks

Family: Translate and operate on written sets • Variant: Union and intersection from listed sets

Method / working

1. List each letter in B or I once, then remove duplicates.
2. For B intersection I', keep letters in B that are not in I: b and z.

Final answer: B union I = {b, r, a, z, i, l, e, n, d}; B intersection I' = {b, z}

$$K = \{k, e, n, y, a\}$$

Cody writes down the statement $B \cap K = \emptyset$
Cody's statement is wrong.

(b) Explain why.

(1)

Worked solution

June 2021 | Question 4 | 1 marks

Family: Translate and operate on written sets • Variant: Complement, universal set and empty set

Method / working

1. The letter a belongs to both B and K, so their intersection contains a.

Final answer: B intersection K is not empty

14 $\mathcal{E} = \{\text{integers } x \text{ such that } 10 \leq x \leq 25\}$

$$A = \{x : x < 18\}$$

$$B = \{x : 13 \leq x < 22\}$$

(a) Write down $n(A)$

(1)

(b) List the members of the set $(A \cup B)'$

(2)

(c) List the members of the set $A' \cap B$

(2)

$$C \subset A, C \subset B \text{ and } n(C) = 5$$

(d) List the members of the set C

(1)

(Total for Question 14 is 6 marks)

Answer reference | May-Nov 2020 | Question 14

Family: Translate and operate on written sets • Variant: Union and intersection from listed sets

14	a		8	1	B1
	b	$A = \{10, 11, 12, 13, 14, 15, 16, 17\}$ $B = \{13, 14, 15, 16, 17, 18, 19, 20, 21\}$ or $A \cup B = \{10, 11, 12, 13, 14, 15, 16, 17, 18, 19, 20, 21\}$			M1 may be seen in a Venn diagram (allow for example 10 – 17 for A and 13 – 21 for B or 10 – 21 for $A \cup B$) or for an answer with one missing element or one extra element
			22, 23, 24, 25	2	A1
	c	$A' = \{18, 19, 20, 21, 22, 23, 24, 25\}$ $B = \{13, 14, 15, 16, 17, 18, 19, 20, 21\}$			M1 may be seen in a Venn diagram (allow 18 – 25 for A' and 13 – 21 for B) or for an answer with one missing element or one extra element
			18, 19, 20, 21	2	A1
	d		13, 14, 15, 16, 17	1	B1
					Total 6 marks

Worked solution

May-Nov 2020 | Question 14 | 6 marks

Family: Translate and operate on written sets • Variant: Union and intersection from listed sets

Official final mark-scheme pages are embedded immediately after this question.

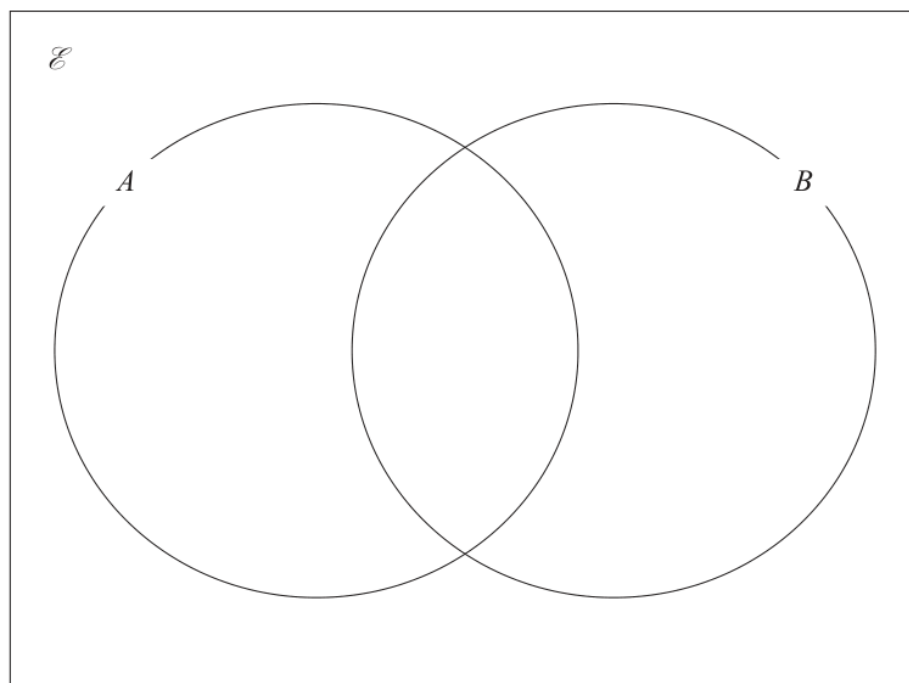
3 $\mathcal{E} = \{11, 12, 13, 14, 15, 16, 17, 18, 19, 20\}$

$$A = \{\text{even numbers}\}$$

$$A \cap B = \{12, 16, 20\}$$

$$(A \cup B)' = \{17, 19\}$$

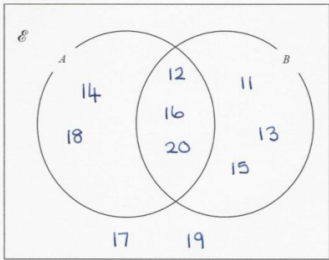
Complete the Venn diagram for the sets $\mathcal{E}$, A and B



(Total for Question 3 is 3 marks)

Answer reference | January 2023 | Question 3

Family: Resolve a three-set Venn model • Variant: Read and combine three-set regions

Question	working	Answer	Mark	Notes
3			3	B3 Fully correct (B2 for 2 or 3 'regions' correct, B1 for one 'region' correct)
				Total 3 marks

Worked solution

January 2023 | Question 3 | 3 marks

Family: Resolve a three-set Venn model • Variant: Read and combine three-set regions

Official final mark-scheme pages are embedded immediately after this question.